

CHAPTER 5 • FOR PHYSIOTHERAPY

When the vessel breaks

Pathophysiology of non-traumatic
intracerebral haemorrhage

Oxford Textbook of Stroke & Cerebrovascular Disease, OUP 2014

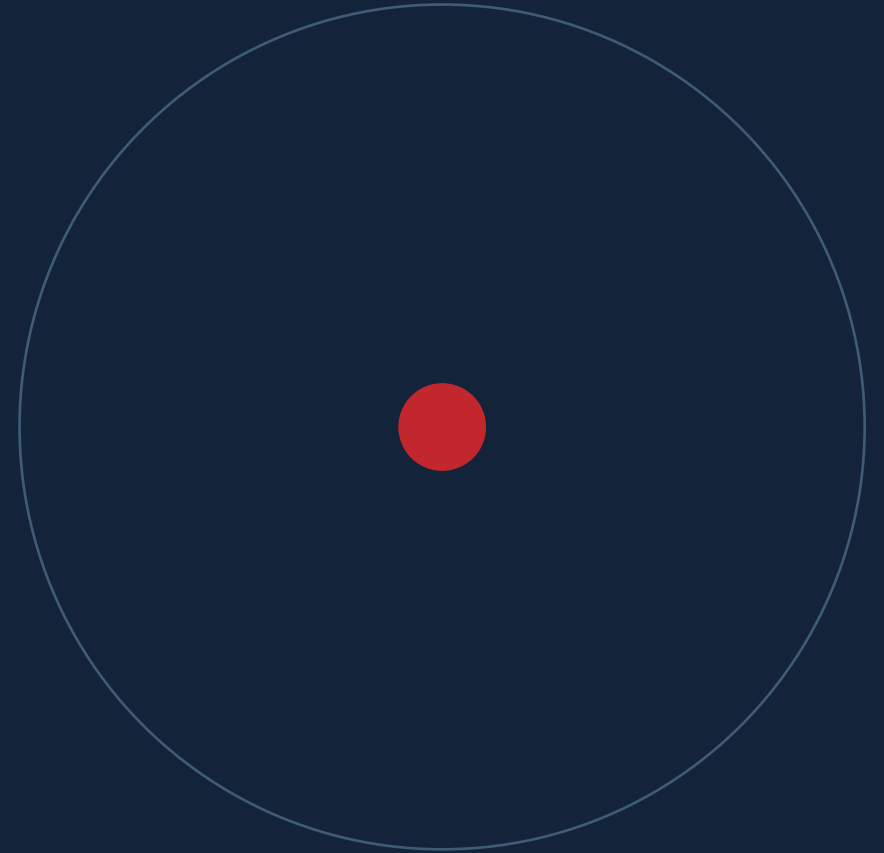
Chapter authors: Constanza Rossi & Charlotte Cordonnier



10 December 1996

7 a.m. · Jill Bolte Taylor, neuroanatomist, aged 37

“I woke up to a pounding pain behind my left eye.”



Source: Jill Bolte Taylor, "My stroke of insight", TED 2008.

10 December 1996

7 a.m. · Jill Bolte Taylor, neuroanatomist, aged 37

Within four hours she could not **walk, talk, read or write.**



Source: Jill Bolte Taylor, "My stroke of insight", TED 2008.

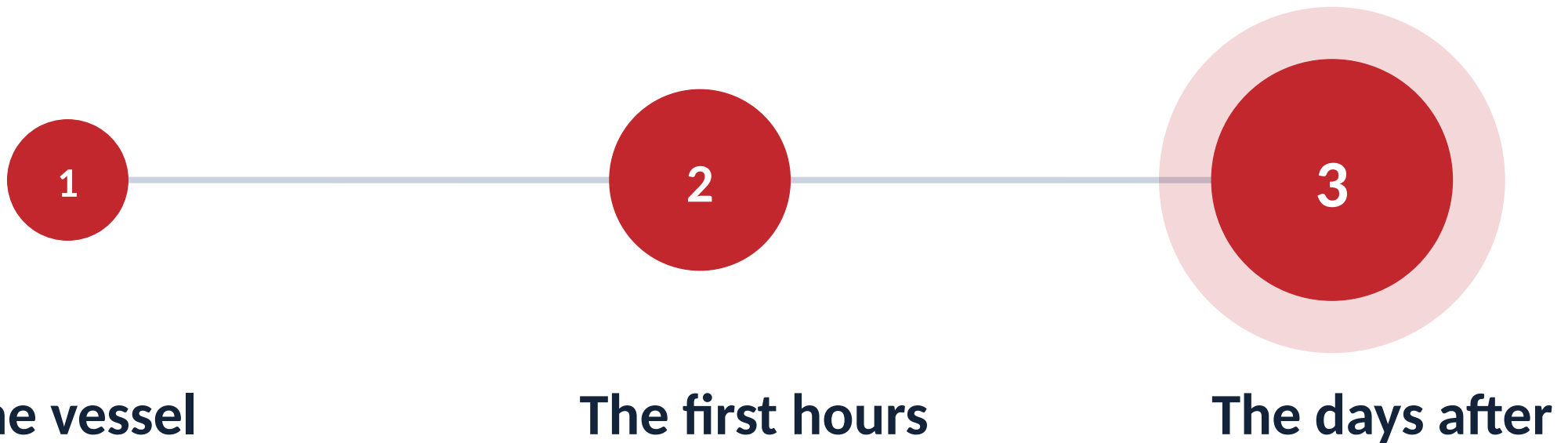
40%

**die within a month of an
intracerebral haemorrhage**

No improvement from 1980 to 2008.

van Asch et al., Lancet Neurol 2010 (36 population studies, 8145 patients): median 1-month case fatality 40.4%.

ICH is a process, not a moment



Each stage is a window to act.

Four things you will be able to do

Explain

why a bleed keeps injuring the brain

Predict

the impairment and the risk of decline

Appraise

the strength of each claim

Apply

time and pace therapy to the stage

Two ways to listen

THE TITLE

is the message. New to stroke?
Hold on to it.

THE TAG

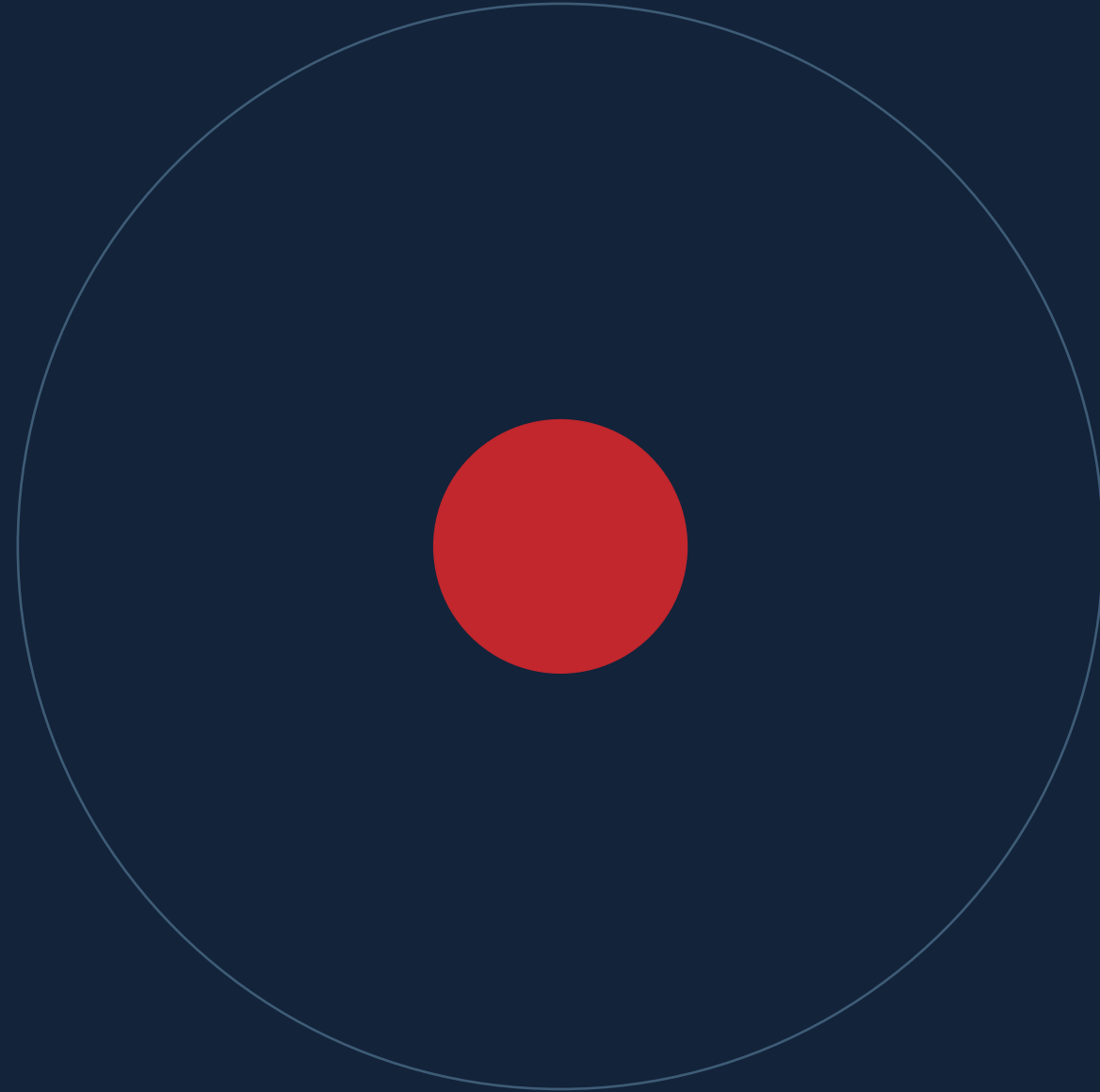
🔍 Cohort • n = 1013

is the evidence. Experienced?
Challenge it.

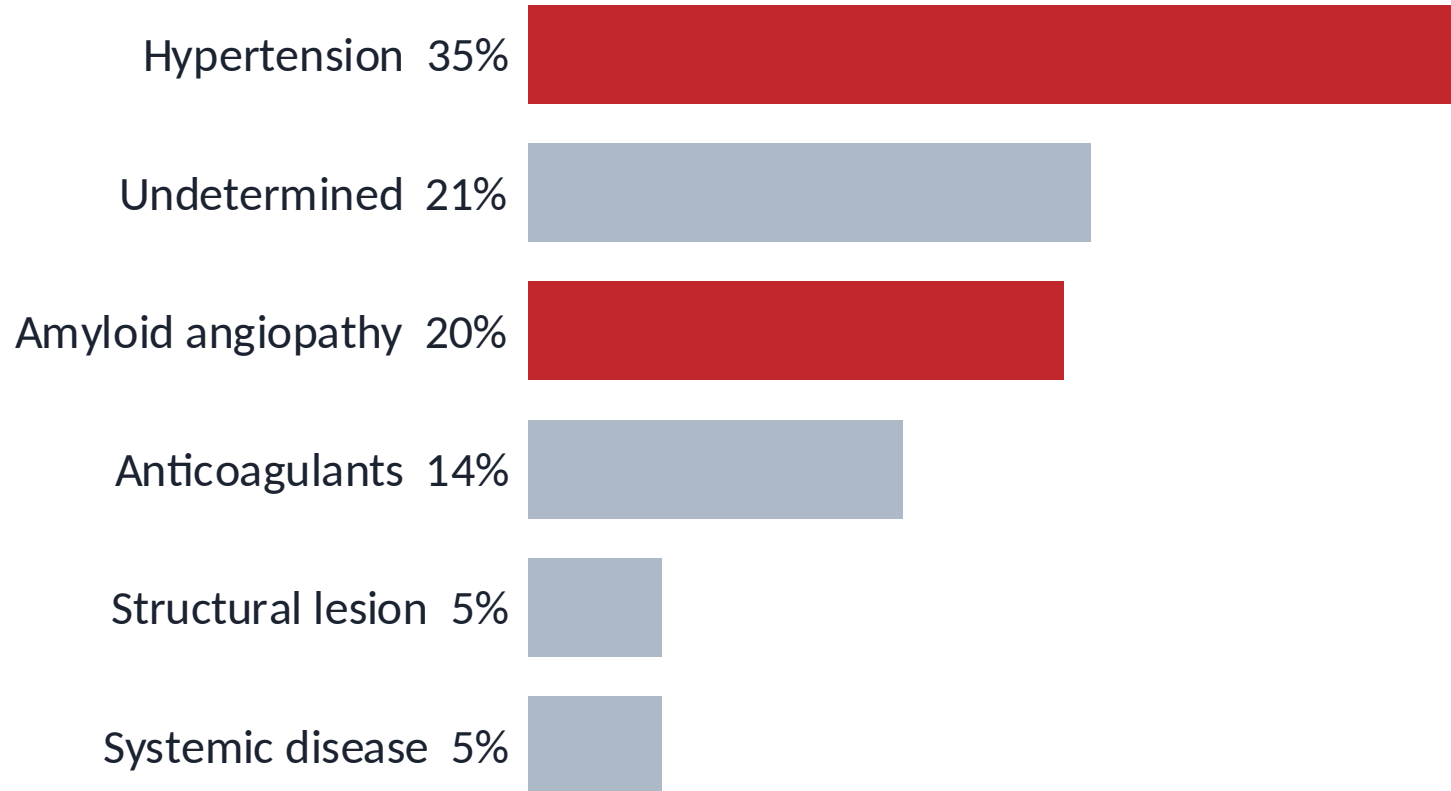
01

The vessel

Why does a small artery give way?



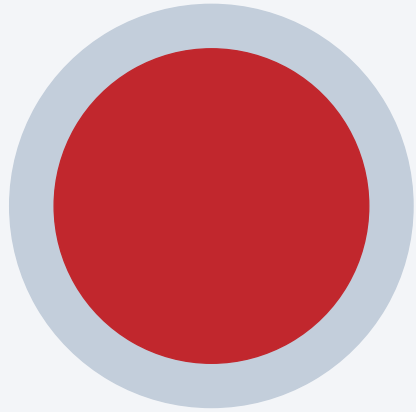
Hypertension and amyloid cause over half



55%
two small-vessel
diseases

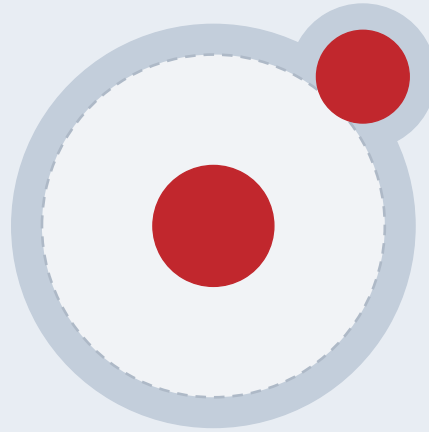
SMASH-U: Meretoja et al., Stroke 2012 (n = 1013, Helsinki). Full breakdown on the handout.

Two diseases, two sets of vessels



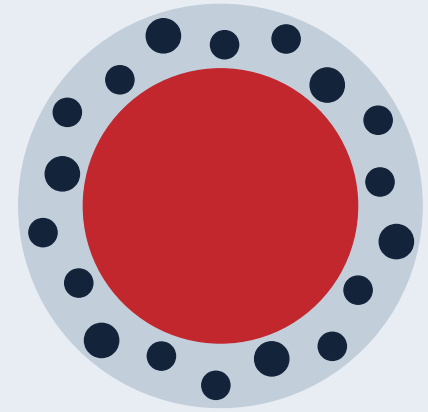
Healthy

Thin wall, wide channel



Hypertension

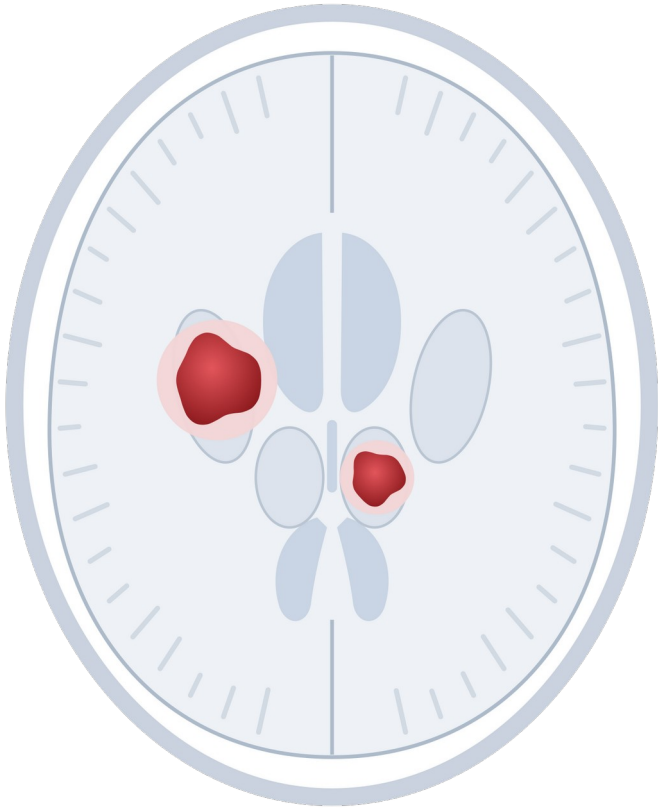
Deep arteries: thick, brittle wall



Amyloid angiopathy

Surface arteries: amyloid replaces muscle

Where it bleeds tells you why



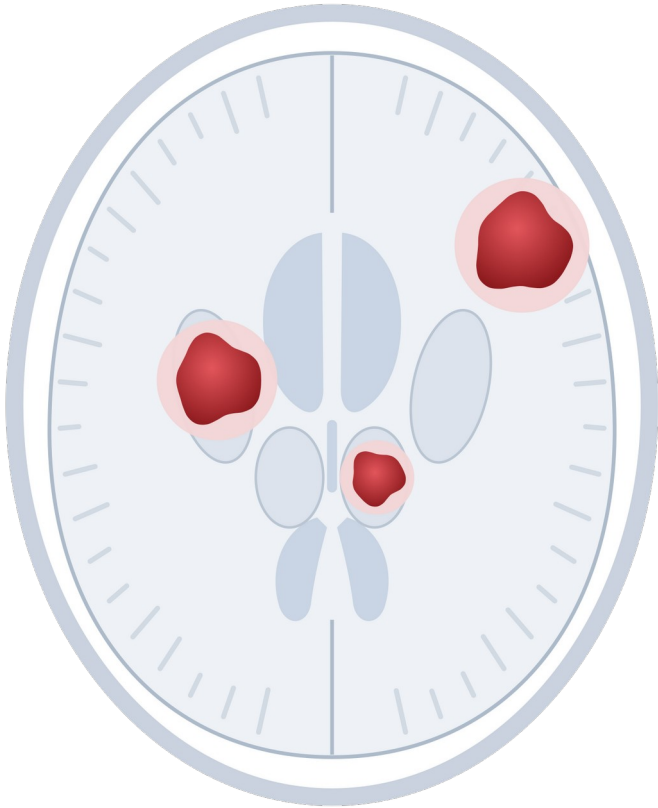
Deep

→ hypertension

putamen • thalamus • pons • cerebellum

Schematic axial slice. The pons and cerebellum sit below this level.

Where it bleeds tells you why



Deep

→ hypertension

putamen • thalamus • pons • cerebellum

Lobar

→ amyloid (older adults)

Young or atypical? Look for a lesion.

Schematic axial slice. The pons and cerebellum sit below this level.

Location predicts the impairment

Putamen, internal capsule Opposite-side weakness. **Clot on the PLIC: worse motor outcome.**

Thalamus Opposite-side sensory loss, ataxia; later, central pain

Cerebellum Same-side ataxia, truncal sway. Drowsy = emergency.

Pons Often severe: four-limb weakness, cranial nerve signs

Lobar By lobe: weakness, neglect, aphasia, field loss; seizures

Microbleeds are the footprints



Surface
→ amyloid



Deep
→ hypertension

60%

**of people with ICH have
microbleeds on MRI**

Healthy adults: 5%

Cordonnier, Al-Shahi Salman & Wardlaw, Brain 2007 (systematic review). Boston criteria on the handout.

Jill: primary or secondary?



37 years old



No hypertension



Lobar bleed



Secondary: an arteriovenous malformation



37 years old



No hypertension



Lobar bleed

65%

of patients like Jill had a lesion on angiography

0%

if over 45, hypertensive, with a deep bleed

Zhu et al., Stroke 1997 (prospective; 206 patients, all had catheter angiography). Other secondary causes on the handout.

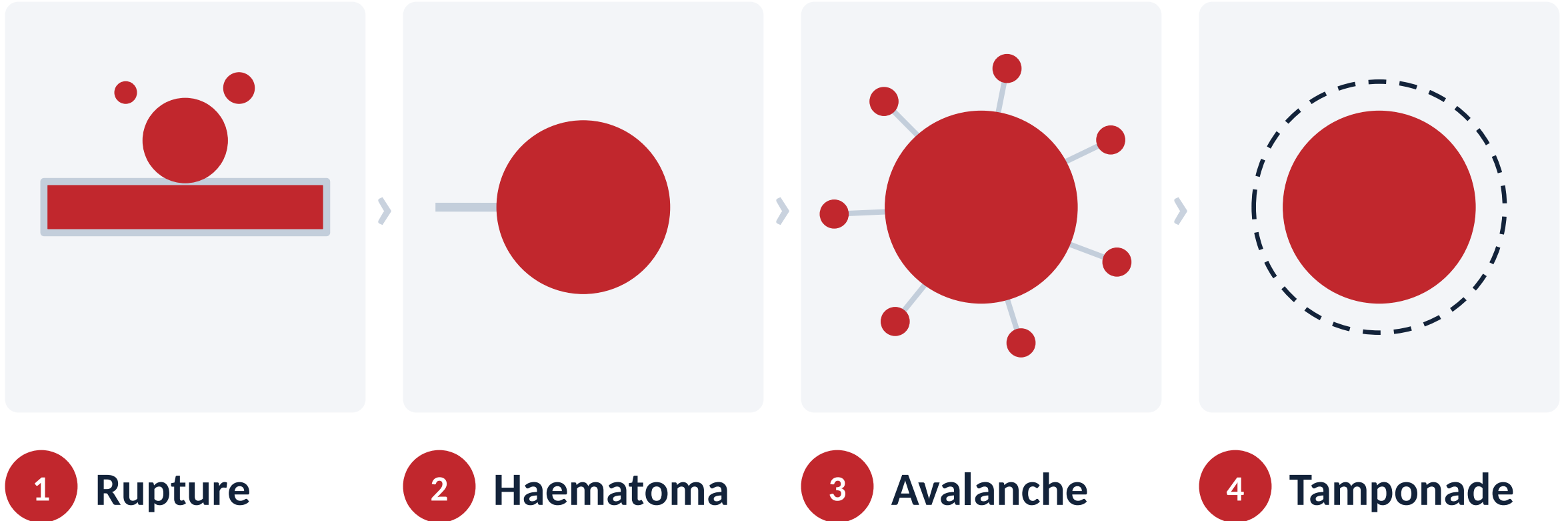
02

The first hours

Rupture, growth and pressure

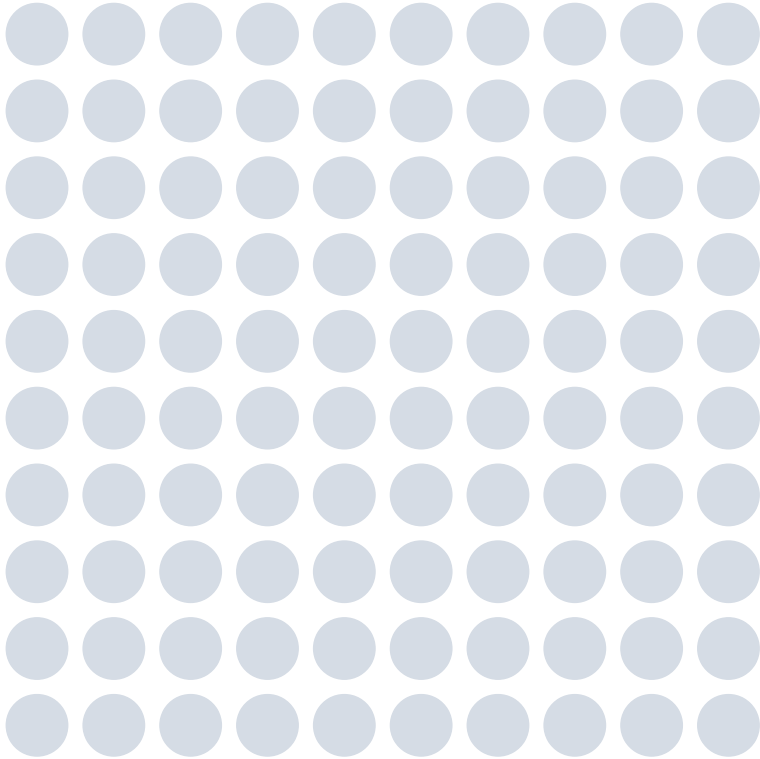


One rupture can trigger more



Fisher CM, J Neuropathol Exp Neurol 1971: secondary bleeding points at the edge of the haematoma.

How many bleeds keep growing?



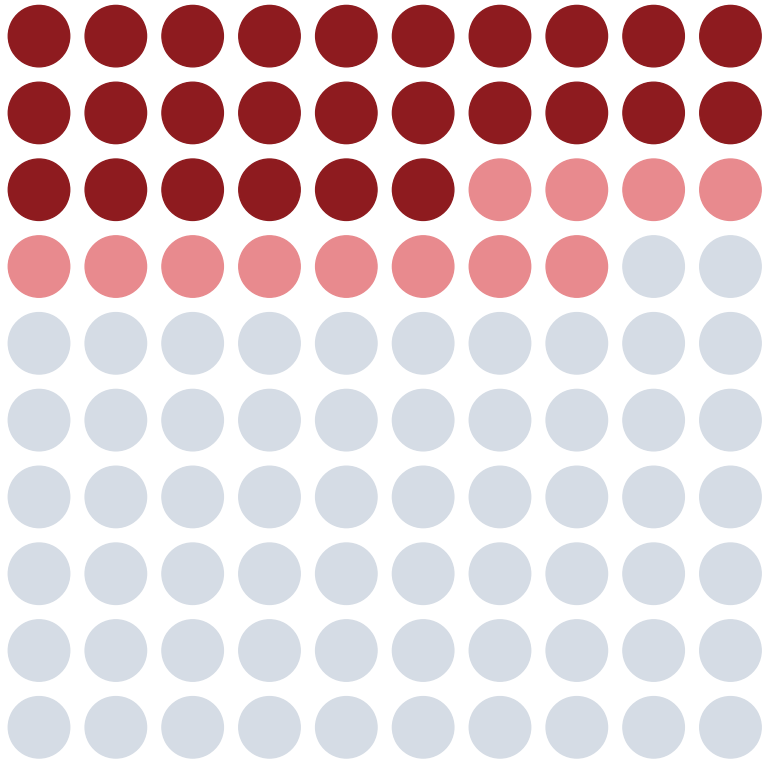
Each dot = one patient, scanned within 3 hours of onset

A 5 in 100

B 15 in 100

C 40 in 100

38 in 100 keep growing



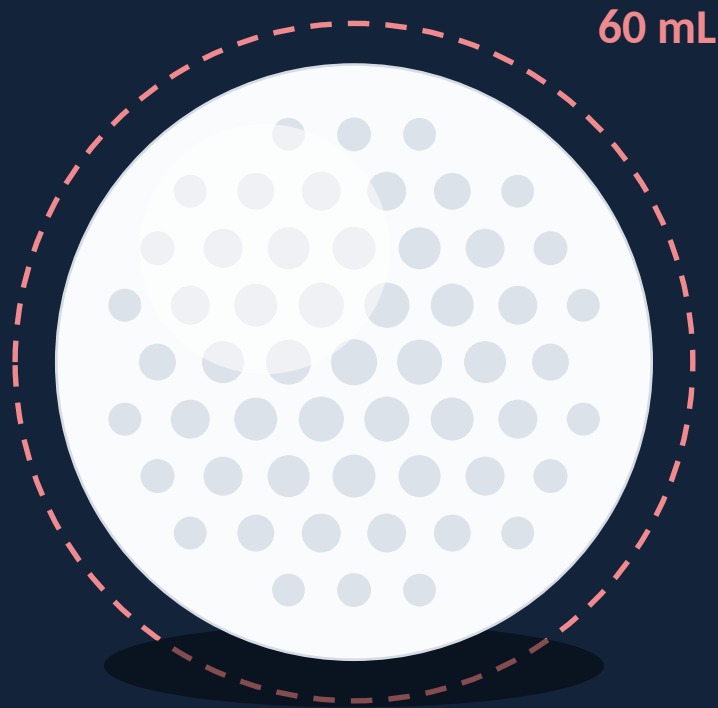
Each dot = one patient, scanned within 3 hours of onset

- 26 grew within the first hour
- 12 more grew within 20 hours
- 62 did not grow substantially

Jill: four hours
from headache to losing her speech

Brott et al., Stroke 1997 (n = 103; growth > 33%). Every 10% of growth: +5% hazard of death (Davis et al., Neurology 2006).

Jill's clot was the size of a golf ball



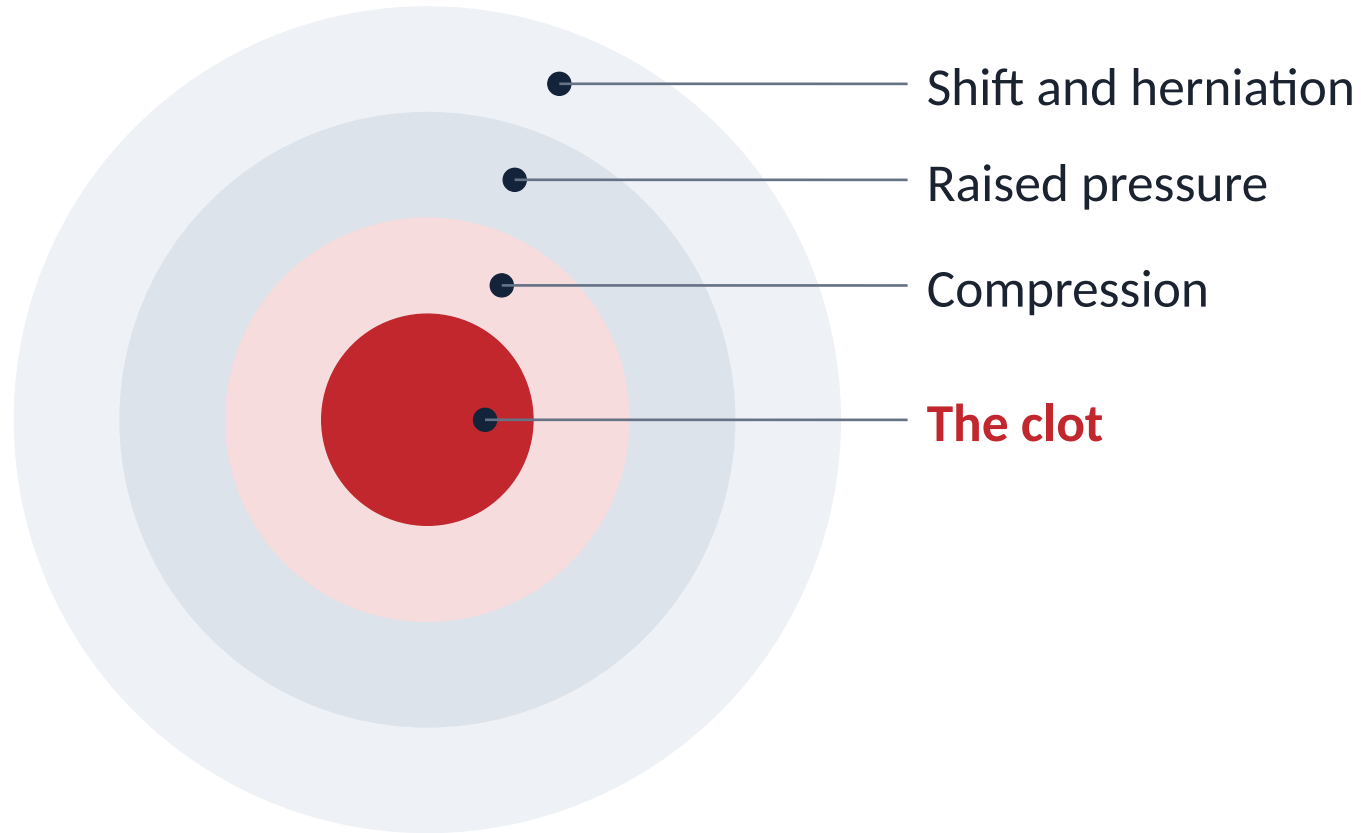
A golf ball $\approx$ 40 mL

60 mL + GCS $\leq$ 8: **91% predicted 30-day mortality**

A clot that grows by a third looks only 10% wider on a scan.

Volume: Broderick et al., Stroke 1993 (Cincinnati, n = 188). A regulation golf ball is 42.7 mm across (about 40.7 mL).

The skull is a closed box



≈45%

break into the ventricles

→ hydrocephalus, worse outcome

Hanley, Stroke 2009.

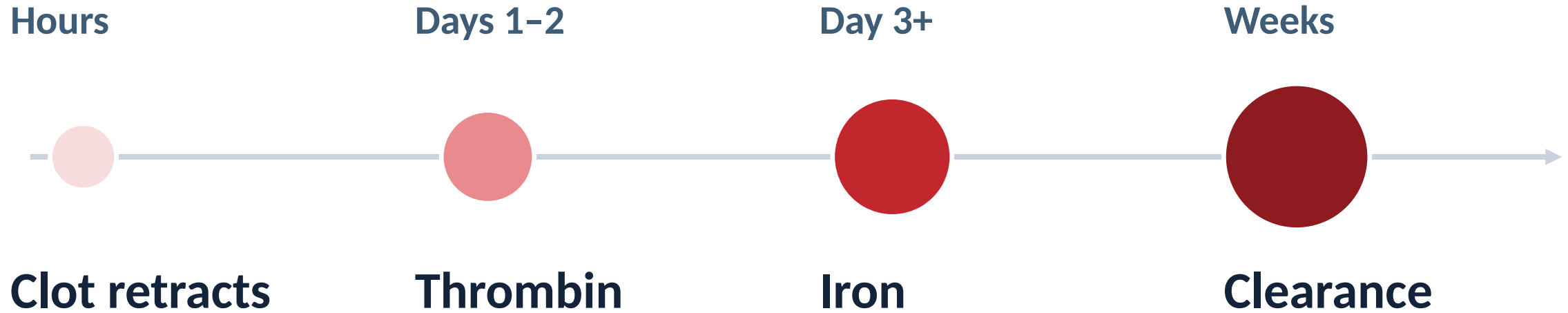
03

The days after

When the blood becomes a toxin



Oedema builds in waves



+75% oedema in the first 24 hours

Jill: surgery 17 days later

Xi, Keep & Hoff, Lancet Neurol 2006; Gebel et al., Stroke 2002.

Three toxins, one result



Thrombin



Iron



Inflammation



Leaky barrier → more oedema → neuronal death

Xi, Keep & Hoff, Lancet Neurol 2006. The deferoxamine trial i-DEF (Selim, Lancet Neurol 2019) was neutral.

True or false?

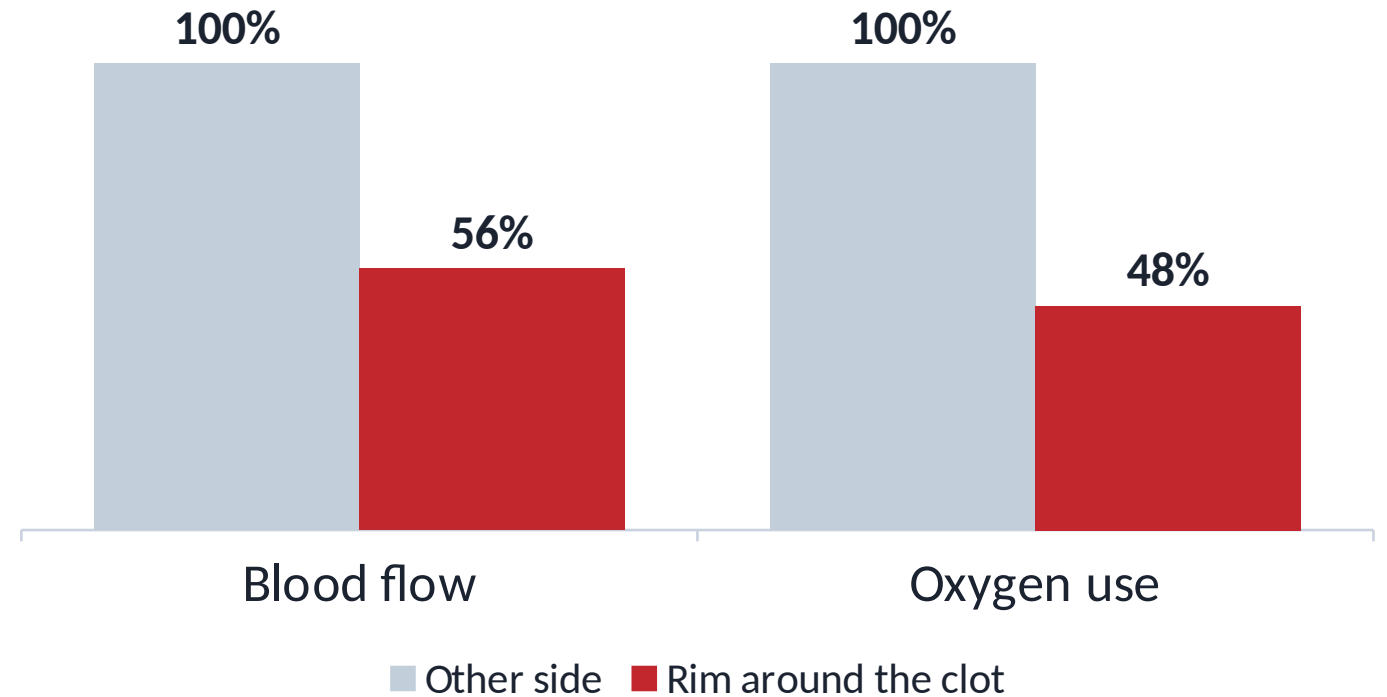
“The rim around the clot is starving tissue.”

False: the rim is resting, not starving

“The rim around the clot is starving tissue.”

Lowering BP below 150 did not reduce flow around the clot (ICH ADAPT).

Rim vs the same area on the other side (PET)



Zazulia et al., J Cereb Blood Flow Metab 2001 (n = 19, 5–22 h after onset); Butcher et al., Stroke 2013.

04

At the bedside

What it means for physiotherapy



Early helps; very early and intense may not

AVERT 2015 • RCT • n = 2104 • infarcts and bleeds

High dose within 24 h: fewer good outcomes (46% vs 50%)

Liu 2014 • RCT • n = 243 • ICH

Rehab started within 48 h: better MBI, fewer deaths

Yen 2019 • RCT • n = 60 • mild to moderate ICH

Out of bed at 24–72 h: better FIM-motor and walking



Short and frequent did better than long sessions (AVERT dose analysis) **Certainty: low (17 RCTs, Kan 2026)**

Bernhardt, Lancet 2015 and Neurology 2016 • Liu, Stroke 2014 • Yen, Neurorehabil Neural Repair 2019 • Kan, Medicine 2026.

Day 4: drowsier than yesterday

MR K, 68 • A COMPOSITE CASE

Left putaminal bleed, 22 mL

Long-standing hypertension

Yesterday: sat out 20 min, GCS 15

TODAY, 9 A.M.

- **GCS 13**
- **New headache**
- **Vomited once**
- **BP 184/98**

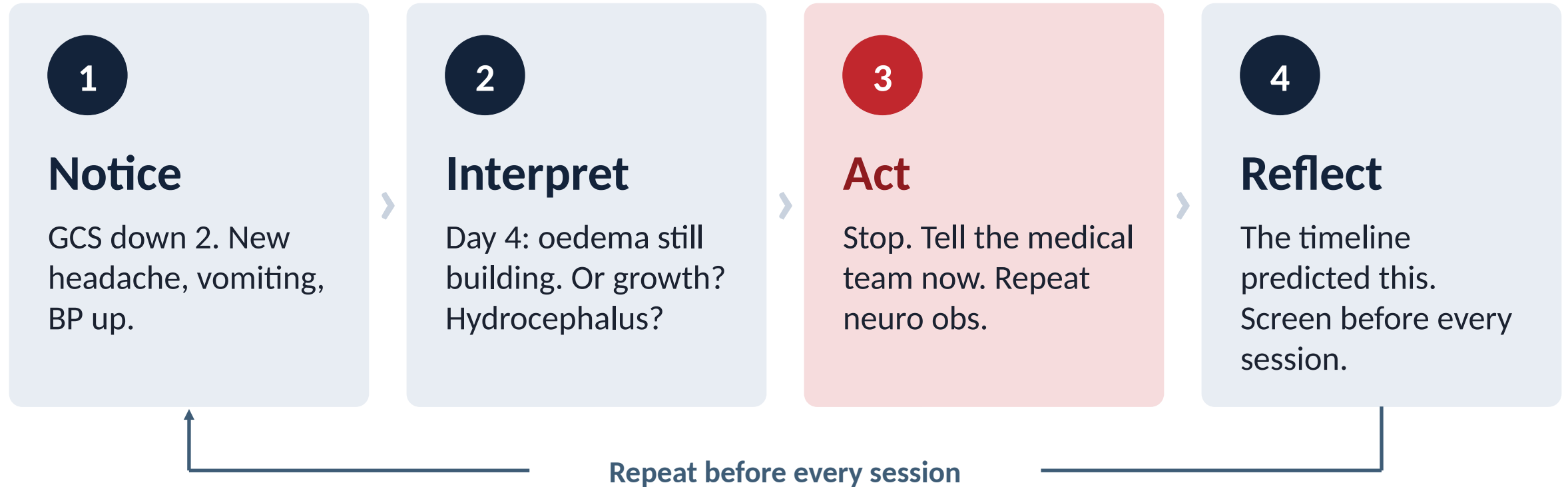
Planned: sit-to-stand practice. What do you do?

A Go ahead, but shorter

B Bed-level work only

C Hold and escalate

C: hold the session and escalate



Reasoning cycle adapted from Levett-Jones et al., Nurse Educ Today 2010. A fall of 2+ GCS points is a common research definition of deterioration.

It depends what you match

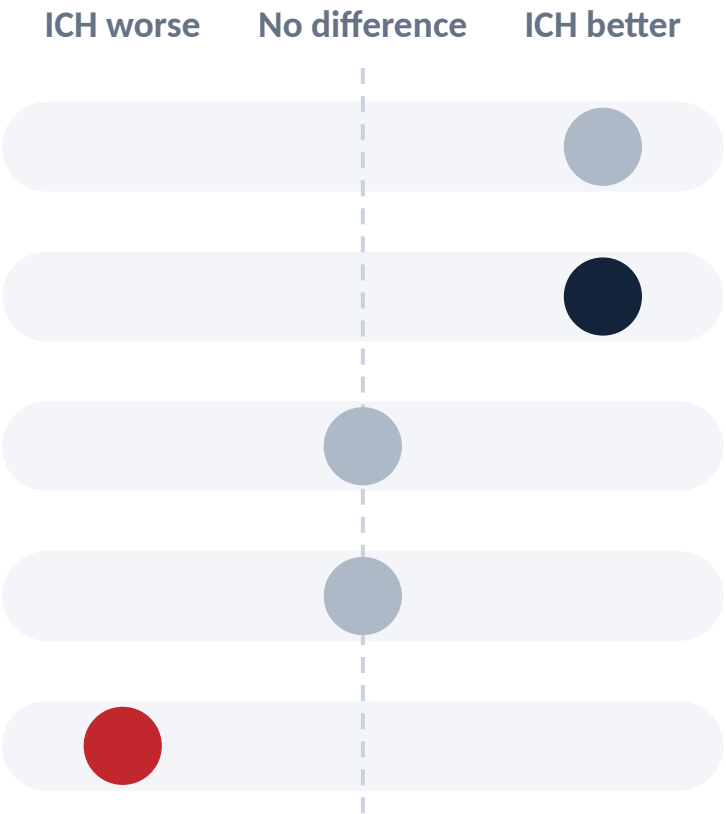
Kelly 2003 • n = 1064
unmatched

Paolucci 2003 • n = 270
matched on the deficit

Salvadori 2020 • n = 229
unmatched

Oosterveer 2022 • n = 360
unmatched

Balk 2026 • n = 194
matched on lesion size and site



MATCH ON THE DEFICIT

ICH looks better

More of the deficit is pressure, and pressure resolves.

MATCH ON THE LESION

ICH looks worse

The blood itself is toxic.

Each dot shows the direction of the main finding, not its size. Full citations on the reference slides.

Every stage is a window to act



The vessel
Before and after

Know the cause. Check BP before, during and after activity.



The first hours
First 24–72 hours

Neuro obs first. Short, gentle sessions; nothing intense.



The days after
Days to weeks

Screen every session. Expect ups and downs. Drowsy: escalate.



The months after
Weeks to months

No early ceiling. Target what the location predicts.

If you remember three things

1

The vessel decides where it bleeds.

2

The first hours decide how big it gets.

3

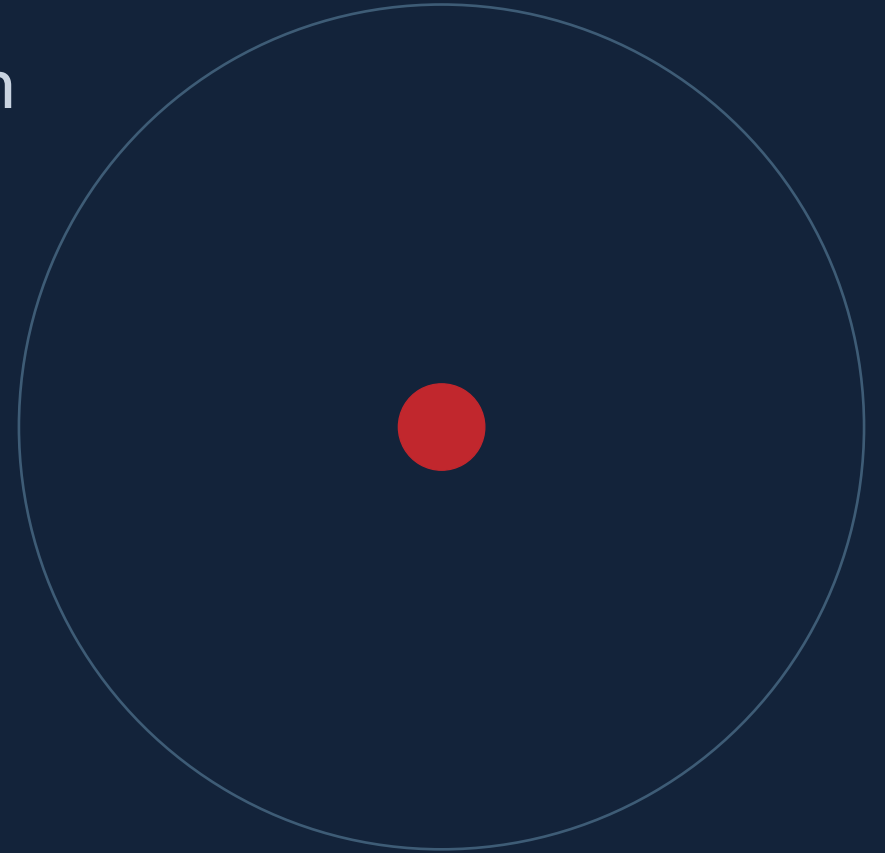
The blood keeps injuring the brain for days.

EIGHT YEARS LATER

Jill walked onto the TED stage holding a human brain.

**ICH is a process,
not a moment.**

Which claim would you challenge?



References: the chapter story

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SOURCES

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